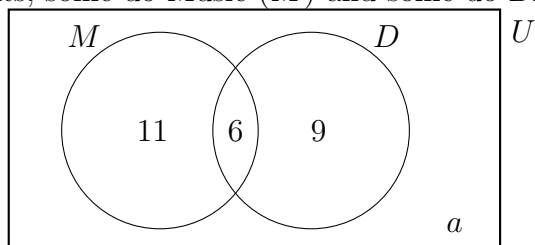


4.19 Quiz: Probability Review — Markscheme

1. In a class of 32 students, some do Music (M) and some do Drama (D).



- (a) Write down the number of students who do Music. [1]

$$n(M) = 11 + 6 = 17 \quad (\mathbf{A1})$$

- (b) Find the number of students who do neither Music nor Drama, a . [1]

$$a = 32 - 11 - 6 - 9 = 6 \quad (\mathbf{A1})$$

- (c) One student is chosen at random. Find the probability that

- (i) the student does not do Music; [1]

$$P(M') = \frac{9 + 6}{32} = \frac{15}{32} \quad (\mathbf{A1})$$

- (ii) the student does Music or Drama, but not both. [2]

$$\text{Identifying the correct regions: } 11 + 9 = 20 \quad (\mathbf{M1})$$

$$P = \frac{20}{32} = \frac{5}{8} \quad (\mathbf{A1})$$

- (d) Given that a student does Drama, find the probability that they also do Music. [2]

$$\text{Recognizing conditional probability, } n(D) = 6 + 9 = 15 \quad (\mathbf{M1})$$

$$P(M \mid D) = \frac{6}{15} = \frac{2}{5} \quad (\mathbf{A1})$$

2. The following table shows whether 16 students are in the IB math program and whether they have a TI nSpire calculator.

	nSpire	No nSpire	Total
IB	8	2	10
Non-IB	1	5	6
Total	9	7	16

- (a) Find the probability that a randomly chosen student is in IB and has a nSpire. [1]

$$P(\text{IB} \cap \text{nSpire}) = \frac{8}{16} = \frac{1}{2} \quad (\text{A1})$$

- (b) Find $P(\text{nSpire} \mid \text{non-IB})$. [2]

$$\text{Identifying non-IB total: } n(\text{non-IB}) = 6 \quad (\text{M1})$$

$$P(\text{nSpire} \mid \text{non-IB}) = \frac{1}{6} \quad (\text{A1})$$

- (c) Determine whether “is in IB” and “has a nSpire” are independent events. Show your working and justify your answer. [3]

$$P(\text{IB}) \times P(\text{nSpire}) = \frac{10}{16} \times \frac{9}{16} = \frac{90}{256} \approx 0.352 \quad (\text{M1A1})$$

$$P(\text{IB} \cap \text{nSpire}) = \frac{8}{16} = 0.5$$

$$0.352 \neq 0.5, \text{ so the events are } \mathbf{not \ independent}. \quad (\text{R1})$$

3. At a school, 60% of students take the bus and 40% walk. Of those who take the bus, 10% arrive late. Of those who walk, 5% arrive late.

- (a) Complete the tree diagram below, showing all probabilities. [2]

Missing values: Walk = 0.4; Bus on time = 0.90; Walk late = 0.05; Walk on time = 0.95 **(A1A1)**

Compound probabilities:

$$\text{Bus} \cap \text{Late}: 0.6 \times 0.10 = 0.06 \quad \text{Bus} \cap \text{On time}: 0.6 \times 0.90 = 0.54$$

$$\text{Walk} \cap \text{Late}: 0.4 \times 0.05 = 0.02 \quad \text{Walk} \cap \text{On time}: 0.4 \times 0.95 = 0.38$$

- (b) Find the probability that a randomly chosen student walks and arrives late. [1]

$$P(\text{Walk} \cap \text{Late}) = 0.4 \times 0.05 = 0.02 \quad \textbf{(A1)}$$

- (c) Find the probability that a randomly chosen student arrives late. [2]

$$P(\text{Late}) = 0.06 + 0.02 \quad \textbf{(M1)}$$

$$= 0.08 \quad \textbf{(A1)}$$

- (d) Given that a student arrives late, find the probability that they walk. [2]

$$P(\text{Walk} \mid \text{Late}) = \frac{P(\text{Walk} \cap \text{Late})}{P(\text{Late})} = \frac{0.02}{0.08} \quad \textbf{(M1)}$$

$$= 0.25 \quad \textbf{(A1)}$$

4. A bag contains 4 black marbles and 6 white marbles. Two marbles are drawn at random, one after the other, without replacement.

- (a) Find the probability that both marbles are black. [2]

$$P(\text{BB}) = \frac{4}{10} \times \frac{3}{9} \quad (\text{M1})$$

$$= \frac{12}{90} = \frac{2}{15} \quad (\text{A1})$$

- (b) Find the probability that both marbles are the same colour. [3]

$$P(\text{WW}) = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3} \quad (\text{M1A1})$$

$$P(\text{same}) = \frac{2}{15} + \frac{1}{3} = \frac{2}{15} + \frac{5}{15} = \frac{7}{15} \quad (\text{A1})$$

- (c) Find the probability that both marbles are black given that at least one marble is black. [3]

$$P(\text{at least one black}) = 1 - P(\text{WW}) = 1 - \frac{1}{3} = \frac{2}{3} \quad (\text{M1A1})$$

$$P(\text{BB} \mid \text{at least one black}) = \frac{2/15}{2/3} = \frac{2}{15} \times \frac{3}{2} = \frac{1}{5} \quad (\text{A1})$$

Total: 28 marks